

Lesson 04(b)

Soil Classification



Topics for the day;

1. Plasticity of soils
2. Classification of fine grained soils using
Atterberg Limits Tests;
 liquid limit test
 plastic limit test
3. Soil classification by Unified Soil Classification
System

Plasticity of Soils

The ability of a soil to undergo unrecoverable or permanent deformation without cracking or crumbling is described as plasticity.

Plasticity is due to the presence of clay and water in a soil.

Soils that cannot be rolled in to a thread at any water content are called non - plastic.

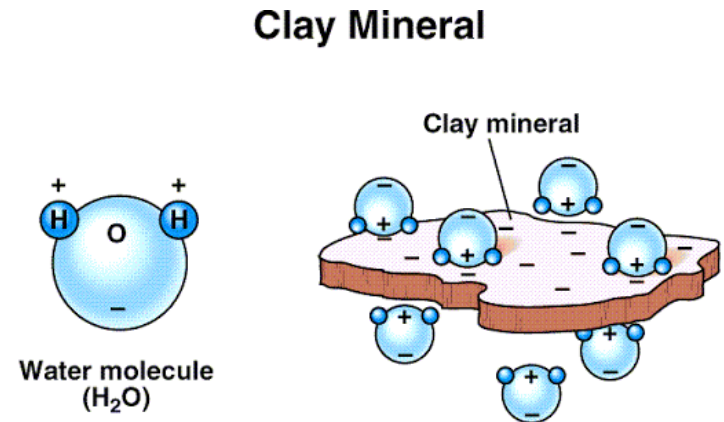


Plasticity of Soilscont.

Negative charges are present on the surfaces of clay particles. The water molecules are dipolar and hence they are attracted towards the surfaces of clay particles. This process is known as adsorption of water.

This adsorbed water contributes to the plasticity of the soil since it allows the clay particles to slip over one another and deform.

These deformations are unrecoverable hence called plastic deformation.



Classification of Fine Grained Soils

Fine soils are classified according to the water contents at which they undergo changes in consistency.



Consistency of fine grain soils varies with the water content of the soil.



The consistency of these soils can range from *solid* (dry) to *semi-solid* to *plastic* to *liquid* (wet).

The water contents at which the consistency changes from one state to the next are called **consistency limits** (or **Atterberg limits**).

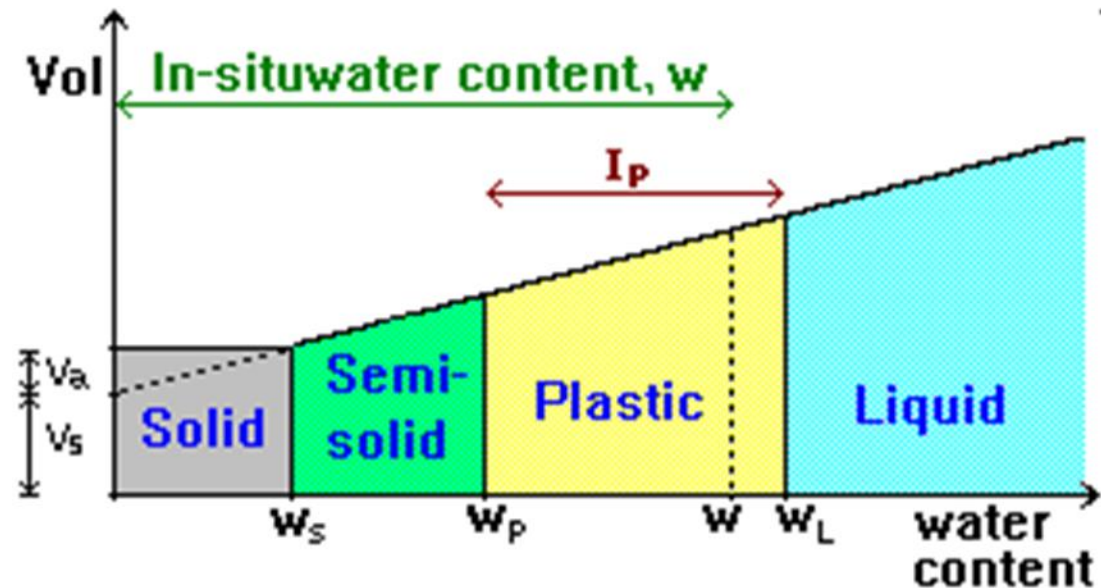
Consistency Limits/ Atterberg Limits of Soils

Consistency Limits represent the water contents at which the soil changes from one state to another state.

Liquid Limit (w_L/LL)

Plastic Limit (w_P/PL)

Shrinkage Limit (w_S/SL)



Liquid Limit

It is the water content at which soil changes from liquid state to plastic state or it is the minimum water content at which soil just begins to flow.

Plastic Limit

It is the water content at which soil changes from plastic state to semi-solid state or it is the minimum water content at which soil rolled into threads of 3 mm diameter just crumbles.

Shrinkage Limit

It is the water content at which the soil changes from semi-solid state to solid state or it is the maximum water content at which further reduction in the water content will not cause decrease in the volume of soil.

The range of water content over which a soil is plastic is termed the **Plasticity Index** (I_p or PI).

Measurements of liquid and plastic limit values can be obtained from laboratory tests.

These tests are called Atterberg Limits Tests after the Swedish engineer Albert Atterberg who developed the method of testing.

Atterberg Limits Tests

1.Determination of the Liquid Limit of Soil

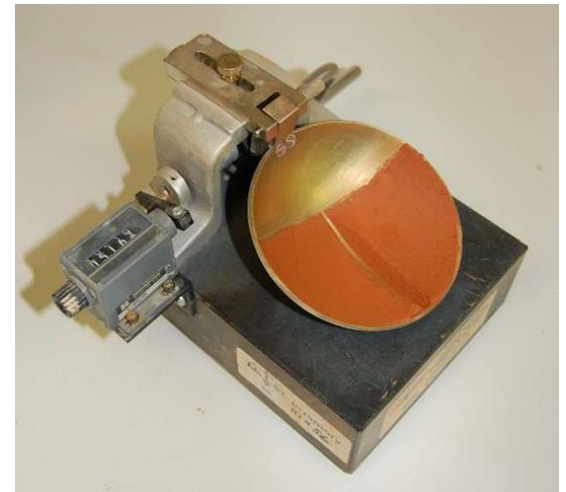
The liquid limit of fine-grained soil is the water content at which soil starts to behave practically like a liquid.

**Liquid limit apparatus
&
Grooving tool**



Procedure

- i) Air-dry the soil sample and break the clods. Remove the organic matter like tree roots, pieces of bark, etc.
- ii) About 100g of the specimen passing through 425 μ m sieve is mixed thoroughly with distilled water in the evaporating dish
- iii) Place a portion of the paste in the cup of the liquid limit device.
- iv) Level the mix so as to have a maximum depth of 1cm.
- v) Draw the grooving tool through the sample along the symmetrical axis of the cup, holding the tool perpendicular to the cup.



vi) After the soil pat has been cut the handle is rotated at the rate of about 2 revolutions per second and the no. of blows counted, till the two parts of the soil sample come into contact for about 10mm length.



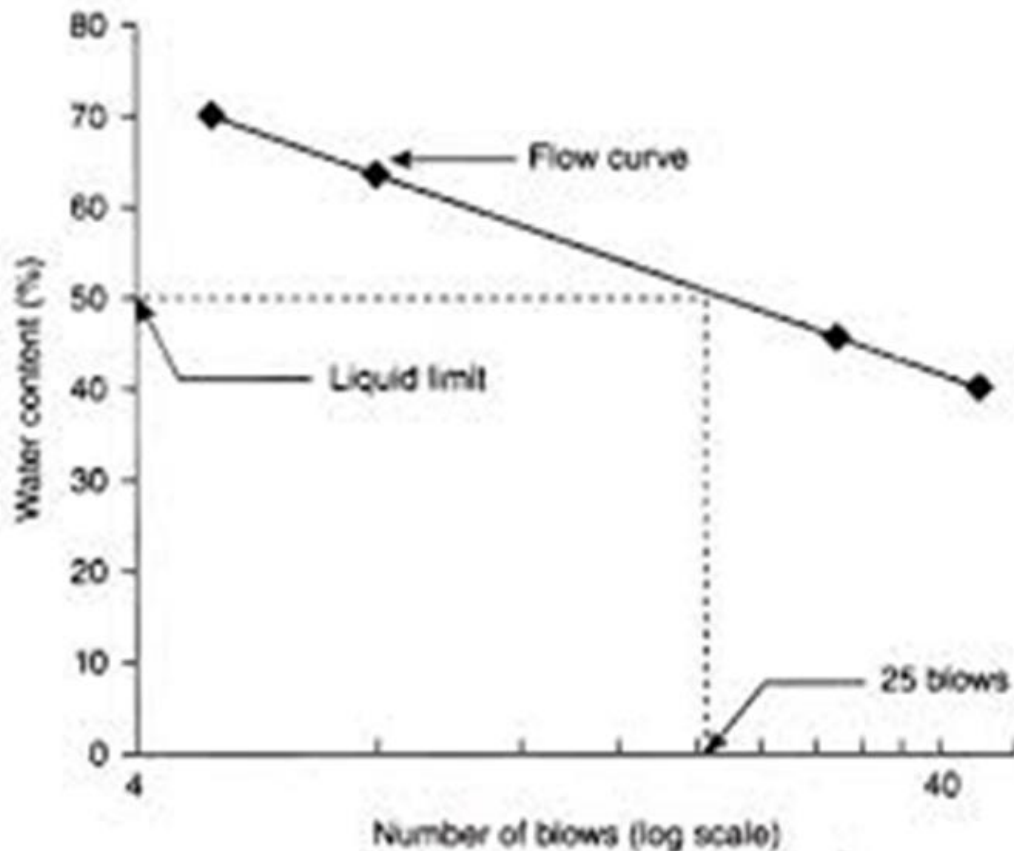
vii) Take about 10g of soil near the closed groove and determine its water content

viii) The soil of the cup is transferred to the dish containing the soil paste and mixed thoroughly after adding a little more water. Repeat the test.

ix) By altering the water content of the soil and repeating the foregoing operations, obtain at least 5 readings in the range of 15 to 50 blows.

Calculations

Liquid limit is determined by plotting a 'flow curve' on a semi-log graph, with **number of blows in x axis (log scale)** and the **water content in y axis** and drawing the best straight line through the plotted points.



Results

The water content corresponding to 25 blows, read from the 'flow curve' is the liquid limit.

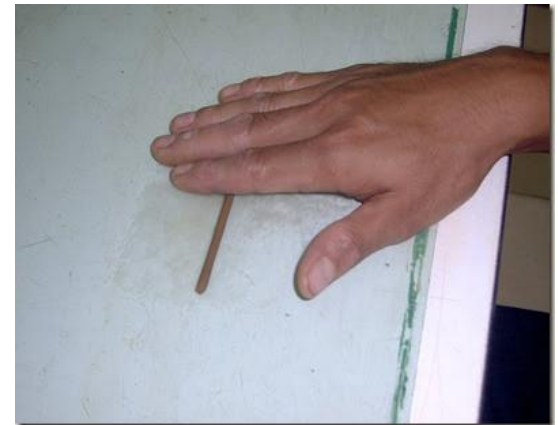
2.Determination of the Plastic Limit of Soil

The plastic limit of fine-grained soil is the water content of the soil below which it ceases to be plastic.



Procedure

- i) Take about 8g of the prepared soil and roll it with fingers on a glass plate.
- ii) If the dia. of the threads can be reduced to less than 3mm, without any cracks appearing, it means that the water content is more than its plastic limit. Knead the soil to reduce the water content and roll it into a thread again.
- iii) Repeat the process of alternate rolling and kneading until the thread crumbles.
- iv) Collect and keep the pieces of crumbled soil thread in the container used to determine the moisture content.
- v) Repeat the process at least twice more with fresh samples of soil each time.



Results

The plastic limit should be determined for at least three portions of the soil sample.

The average water content to the nearest whole number is recorded as the plastic limit.

Plastic Limit test
Observations

Trial number	Moisture can no	Mass of can (g)	Mass of wet soil +can (g)	Mass of dry soil + can (g)
01	A -2	9.5	10.6	10.4
02	A -3	9.6	10.8	10.6

Determine the Plastic Limit.

Determination of Moisture content

Trial No 01

$$\text{Water Content} = \frac{\text{mass of water}}{\text{mass of solid}}$$

$$\text{Water content} = (10.6 - 10.4) / (10.4 - 9.5) = 22.2\%$$

Trial No 02

$$\text{Water content} = (10.8 - 10.6) / (10.6 - 9.6) = 20\%$$

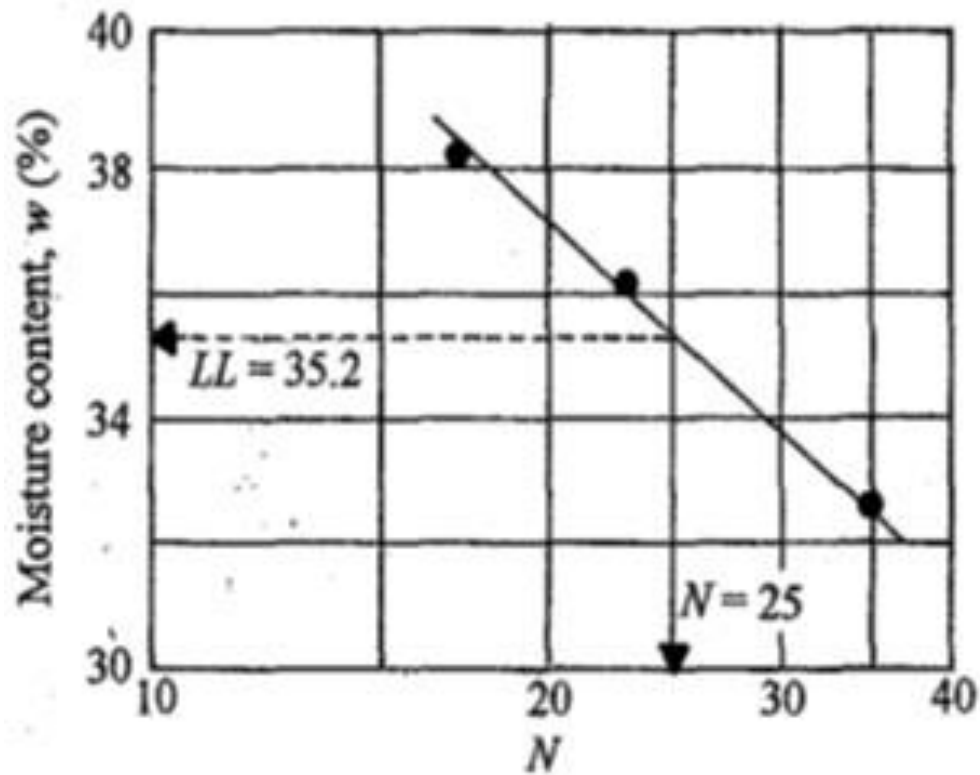
$$\text{Plastic Limit(PL)} = (22.2 + 20)/2 = 21.1\%$$

Liquid Limit Test

Observations

Trial number	Moisture can no	Mass of can (g)	Mass of wet soil + Can (g)	Mass of Dry soil + Can (g)	No of blows
01	M-01	12.3	16.4	15.6	35
03	M-04	11.9	16.3	15.4	23
04	M-05	12.2	16.0	15.2	17

Determine the liquid limit.



Liquid limit(LL) = 35.2%

Plasticity Index = $LL - PL = 35.2\% - 21.1\% = 14.1\%$

Description of soils using plasticity

Liquid Limit	Description	Notation
$w_L = < 35\%$	Low plasticity	L
$w_L = 35 - 50\%$	Intermediate plasticity	I
$w_L = 50 - 70\%$	High plasticity	H
$w_L = 70 - 90\%$	Very high plasticity	V
$w_L = > 90\%$	Extremely high plasticity	E

Soil Classification

Soil scientists use a naming system, or classification to describe and identify soils. Over time different countries have developed different kinds of systems.

Soil Classification Systems

USCS – Unified Soil
Classification System

CSA – Canadian
Standards Association

European Standards
(Eurostandards)

AASHTO – American
Association of State
Highway and
Transportation Officials

ASTM – American
Society for Testing and
Materials

MIT System of soil
classification

Unified Soil Classification System

The **Unified Soil Classification System (USCS)** is used in engineering and geology to describe the texture and grain size of a soil.

The classification system can be applied to most soils, and is represented by a two-letter symbol.

Eg. GW, SP, ML, CH

First letter

Symbol

Definition

G

gravel

S

sand

M

silt

C

clay

O

organic

Second letter

P

poorly graded

W

well graded

H

high plasticity

L

low plasticity

Unified Soil Classification System and Symbol Chart:

Unified soil classification system for coarse grained soils

COARSE-GRAINED SOILS (more than 50% of material is larger than No. 200 sieve size.)		
Clean Gravels (Less than 5% fines)		
GRAVELS More than 50% of coarse fraction larger than No. 4 sieve size		GW Well-graded gravels, gravel-sand mixtures, little or no fines
		GP Poorly-graded gravels, gravel-sand mixtures, little or no fines
	Gravels with fines (More than 12% fines)	
		GM Silty gravels, gravel-sand-silt mixtures
		GC Clayey gravels, gravel-sand-clay mixtures
Clean Sands (Less than 5% fines)		
SANDS 50% or more of coarse fraction smaller than No. 4 sieve size		SW Well-graded sands, gravelly sands, little or no fines
		SP Poorly graded sands, gravelly sands, little or no fines
	Sands with fines (More than 12% fines)	
		SM Silty sands, sand-silt mixtures
		SC Clayey sands, sand-clay mixtures

Sieve Size

No 4 = 4.75mm

No 200 = 0.075mm

Unified soil classification system for fine grained soils

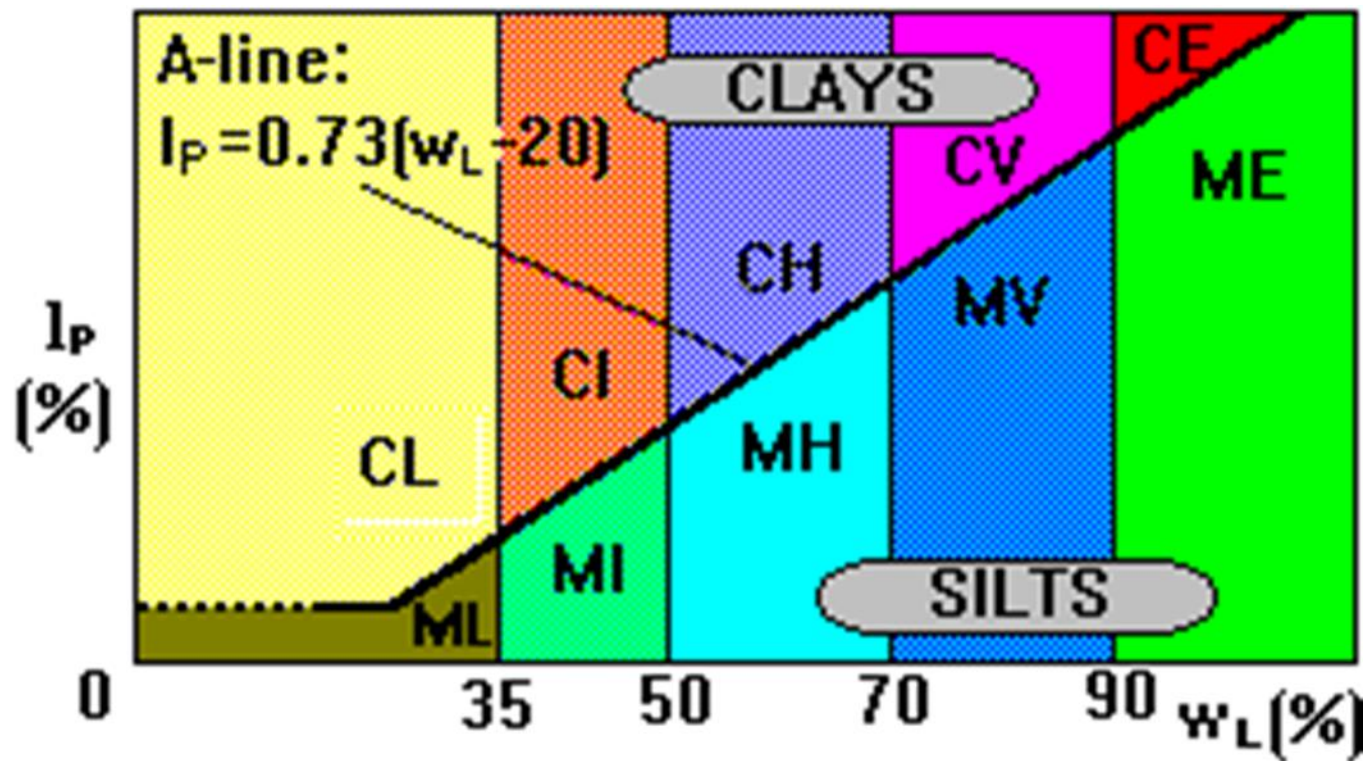
FINE-GRAINED SOILS (50% or more of material is smaller than No. 200 sieve size.)			
SILTS AND CLAYS Liquid limit less than 50%		ML	Inorganic silts and very fine sands, rock flour, silty or clayey fine sands or clayey silts with slight plasticity
		CL	Inorganic clays of low to medium plasticity, gravelly clays, sandy clays, silty clays, lean clays
		OL	Organic silts and organic silty clays of low plasticity
SILTS AND CLAYS Liquid limit 50% or greater		MH	Inorganic silts, micaceous or diatomaceous fine sandy or silty soils, elastic silts
		CH	Inorganic clays of high plasticity, fat clays
		OH	Organic clays of medium to high plasticity, organic silts
HIGHLY ORGANIC SOILS		PT	Peat and other highly organic soils

Sieve Size

No 4 = 4.75mm

No 200 = 0.075mm

Soil Plasticity Chart as per Unified Soil Classification System



Summary:

1. Plasticity of a soil is due to the presence of clay and water.
2. Fine grained soils are classified using the water contents at which they under go changes in consistency.
3. Fine grained soils will change its consistency from solid - semi solid -plastic to liquid with increasing water content.
4. Liquid limit, plastic limit and plasticity index of a soil are determined using Atterberg Limits tests.
5. The grading curve and the consistency limits are used to classify / separate soils in to groups.